

Guía 1: Dinámica no lineal (ODE)

1) Sistemas dinámicos:
$$\begin{cases} \dot{x}_1 = \frac{dx_1}{dt} = f_1(x_1, \dots, x_n) \\ \vdots \\ \dot{x}_n = \frac{dx_n}{dt} = f_n(x_1, \dots, x_n) \end{cases}$$

2) Espacio de fases ($\mathbb{R}^n$): donde viven $x_1, \dots, x_n$

↳ trayectoria: curva parametrizada por t

↳ retrato (diagrama) de fases: un racimo de trayectorias

3) Punto fijo: cero del $\vec{F}(\vec{x}) = (f_1(\vec{x}), \dots, f_n(\vec{x}))$, $\vec{x} = (x_1, \dots, x_n)$

Ej 1.b) $\dot{x} = 4x^2 - 16 = 4(x-2)(x+2) \rightarrow \text{PF: } x_{1,2} = \pm 2$

(resolvemos el sistema)

$$\frac{dx}{dt'} = 4(x-2)(x+2)$$

$$\frac{1}{(x-2)(x+2)} \frac{dx}{dt'} \cdot dt' = 4 dt'$$

$$\int_0^t \frac{1}{(x-2)(x+2)} \frac{dx}{dt'} dt' = \int_0^t 4 dt' = 4t' \Big|_0^t$$

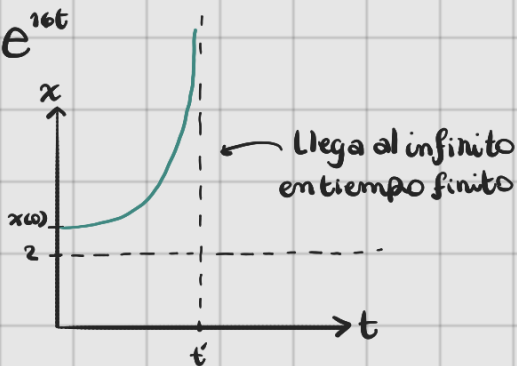
$$\int_{x(0)}^{x(t)} \frac{1}{(x-2)(x+2)} dx = 4t \iff 4t = \frac{1}{4} \left[\ln \left| \frac{x(t)-2}{x(0)-2} \right| - \ln \left| \frac{x(t)+2}{x(0)+2} \right| \right]$$

fracciones simples

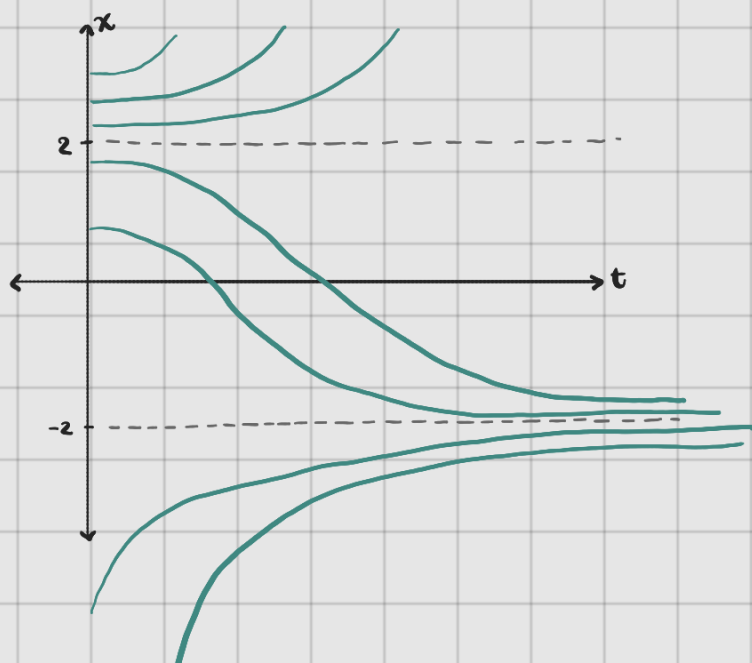
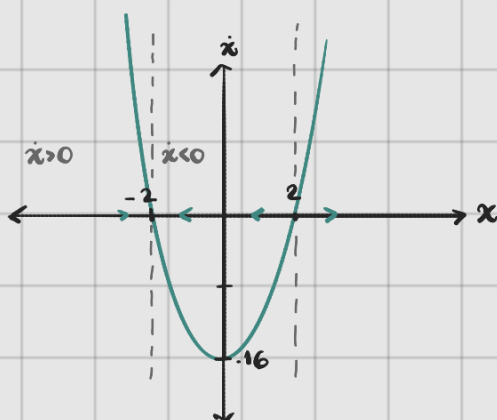
$$\Rightarrow \left| \frac{x(t)-2}{x(t)+2} \right| = \left| \frac{x(0)-2}{x(0)+2} \right| e^{16t}$$

$$\Rightarrow x(t) = 2 \left(\frac{1 + A e^{16t}}{1 - A e^{16t}} \right), \quad |x| > 2$$

$$\hookrightarrow A := \frac{x(0)-2}{x(0)+2} > 0$$



Gráficamente $\dot{x} = 4(x-2)(x+2)$



(soluciones variando $x(0)$)

Retrato de fases



Estabilidad Análisis lineal

$$x(t) = x^* + \overset{\text{perturbación}}{z(t)}$$

$$\frac{dz}{dt} = \frac{d(x-x^*)}{dt} = f(x) - f(x^*) = f(x) - \overset{(\text{Taylor?})}{f(x^* + z(t))} \approx \left. \frac{df}{dx} \right|_{x^*} z(t)$$

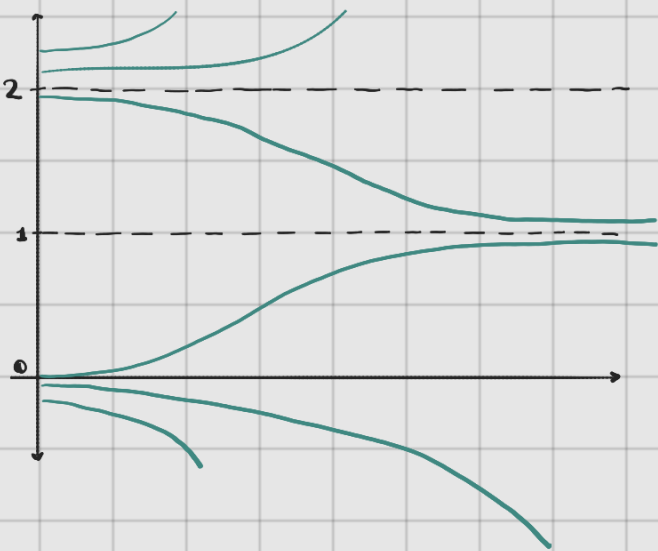
$$z(t) \approx z e^{\left. \frac{df}{dx} \right|_{x^*} t} \begin{cases} \text{Estabilidad: } \left. \frac{df}{dx} \right|_{x^*} < 0 \\ \text{Inestabilidad: } \left. \frac{df}{dx} \right|_{x^*} > 0 \end{cases}$$

$$\Leftrightarrow \tau \approx \left(\left. \frac{df}{dx} \right|_{x^*} \right)^{-1}$$

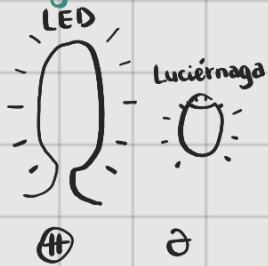
$$\dot{x} = x(x-1)(x-2) = 2x - 3x^2 + x^3 = f(x) \Rightarrow x^* \in \{0, 1, 2\}$$

Estabilidad lineal $\frac{df}{dx} = 2 - 6x + 3x^2 \Rightarrow \begin{cases} \left. \frac{df}{dx} \right|_0 = 2 > 0 \rightarrow \text{Inestable} \\ \left. \frac{df}{dx} \right|_1 = -1 < 0 \rightarrow \text{Estable} \\ \left. \frac{df}{dx} \right|_2 = 2 > 0 \rightarrow \text{Inestable} \end{cases}$

Retrato de fases



Ejercicio de la luciérnaga



$\theta \leftarrow$ Fase de la luciérnaga

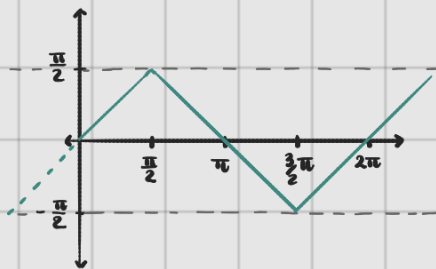
$\phi \leftarrow$ Fase del LED

$$\omega(t) = \Omega t \Rightarrow \dot{\omega} = \Omega$$

$$\dot{\theta} = \omega + A f(\omega - \theta)$$

2π -periódica

$$\Rightarrow f(\phi) = \begin{cases} \phi & \text{si } |\phi| < \frac{\pi}{2} \\ \pi - \phi & \text{si no} \end{cases}$$



$$\Leftrightarrow \phi = \omega - \theta \Rightarrow \dot{\phi} = \dot{\omega} - \dot{\theta} = (\Omega - \omega) - A f(\phi)$$

Diferencia de fase

$\hookrightarrow$ Busco solución tal que $\phi = \text{cte}$ ("están sincronizados")

Es decir, busco PF del sistema

$$\dot{\phi} = 0 \Leftrightarrow \Omega - \omega - A f(\phi) = 0 \Leftrightarrow f(\phi) = \frac{\Omega - \omega}{A} \stackrel{\text{rec.}}{\Rightarrow} |f(\phi^*)| \leq \frac{\pi}{2}$$

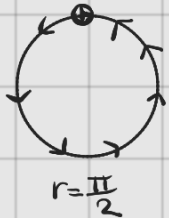
$$\exists \phi^* \text{ PF } \Leftrightarrow \left| \frac{\Omega - \omega}{A} \right| \leq \frac{\pi}{2}$$

$$\text{def } r := \frac{\Omega - \omega}{A}$$

$\emptyset$



$r > \frac{\pi}{2} \Rightarrow \Omega > \frac{A\pi}{2} + \omega \Rightarrow$ La luciernaga siempre tiene menor frec que el LED, nunca lo alcanza (siempre va quedando atrás)



$$r = \frac{\pi}{2}$$



$$r < \frac{\pi}{2}$$

Bifurcaciones

P1.6) $\dot{x} = r x - \frac{x}{1+x} \quad (r > 0)$

PF: $\dot{x} = 0 \Leftrightarrow x \left[r - \frac{1}{1+x} \right] = 0$

$\nearrow x^* = 0$
 $\searrow x^* = \frac{1}{r} - 1$

(pista: si $r=1$, hay un solo PF, luego hay 2 PF en $r < 1$ y $r > 1$, y su estabilidad se invierte luego de cruzar $r=1 \Rightarrow$ se trata de una bifurcación transcritical)

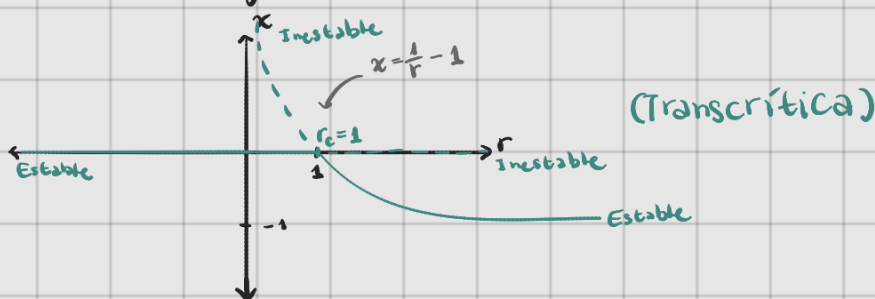
Estabilidad: $\frac{df}{dx} = r - \frac{1}{(1+x)^2}$

$$\left. \frac{df}{dx} \right|_{x^*=0} = r - 1 \Rightarrow \begin{cases} \text{Estable si } r < 1 \\ \text{Inestable si } r > 1 \end{cases}$$

$$\left. \frac{df}{dx} \right|_{x^*=\frac{1}{r}-1} = r - r^2 \Rightarrow \begin{cases} \text{Estable si } r > 1 \\ \text{Inestable si } r < 1 \end{cases}$$

$\Rightarrow$ En $r=r_c=1$, 2 PF se cruzan e intercambian estabilidad

Diagrama de bifurcaciones



P1.8) "Switch bioquímico"

$$\dot{g} = \frac{dg}{dt} = \overset{\text{"source"}}{k_1 S_0} - k_2 g + \frac{k_3 g^2}{k_4 + g^2} \quad (k_j > 0 \forall j=1,2,3,4)$$

Proponemos un cambio de variables $\begin{cases} g = g_0 x \\ t = t_0 \tau \end{cases}$

Luego $\frac{dg}{dt} = \frac{d(g_0 x)}{d(t_0 \tau)} = \frac{g_0}{t_0} \frac{dx}{d\tau} = k_1 S_0 - k_2 g_0 x + \frac{k_3 g_0^2 x^2}{k_4 + g_0^2 x^2}$

(enunciado)

Ver que $\frac{dx}{dz} = s - rx + \frac{x^2}{1+x^2}$

Tenemos que $\frac{dx}{dz} = \kappa_1 \frac{s_0 t_0}{g_0} - \kappa_2 t_0 x + \underbrace{\frac{\kappa_3 t_0 g_0 x^2}{\kappa_4^2 + g_0^2 x^2}}_{\frac{g_0 (\kappa_3 t_0) (x^2)}{g_0^2 (\frac{\kappa_4^2}{g_0^2} + x^2)}} = \kappa_1 \frac{s_0 t_0}{g_0} - \kappa_2 t_0 x + \frac{\kappa_3 t_0}{g_0} \frac{x^2}{(\frac{\kappa_4^2}{g_0^2} + x^2)}$

Pido que $\kappa_4 = g_0 \wedge \kappa_3 t_0 = g_0 \Rightarrow g_0 = \kappa_4$ y luego $t_0 = \frac{\kappa_4}{\kappa_3} \Rightarrow \frac{g_0}{t_0} = \kappa_3$

¿Porque?

Para que el término $\frac{\kappa_3 t_0}{g_0} \frac{x^2}{(\frac{\kappa_4^2}{g_0^2} + x^2)} = \frac{x^2}{1+x^2}$

$\Rightarrow \frac{dx}{dz} = \underbrace{\frac{\kappa_1 s_0}{\kappa_3}}_{=s} - \underbrace{\frac{\kappa_4 \kappa_2}{\kappa_3}}_{=r} x + \frac{x^2}{1+x^2} = s - rx + \frac{x^2}{1+x^2}$